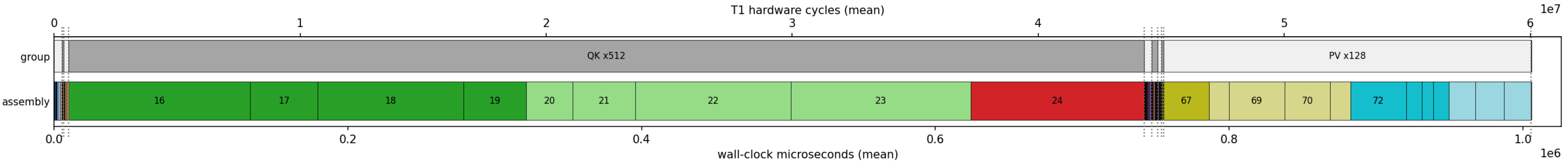


attention_self (ViT patch self-attention): per-instruction breakdown 80 timed rows + 0 dispatcher-only CSR flips; QK body aggregates 512 repeats, PV body 128
total T1 cycles = 60060.2k total wall = 1,006,085us libt1 sliver = 5,081us (@60MHz)



stage	assembly	group	cycles	wall	stage	assembly	group	cycles	wall	stage	assembly	group	cycles	wall	stage	assembly	group	cycles	wall
0	im2col vle8 idxH1 x2	im2col	27.8k	465us	16	QK vrgather.vx K[j] x512	QK x512	7392.2k	123,617us	39	nw vnsrl idx8 x2	Newton	9.9k	166us	64	pv vle8 Va x2	PV pre	28.3k	474us
1	im2col vle8 idxV x2	im2col	27.7k	464us	17	QK vwmulu.vv x512	QK x512	2739.0k	46,065us	40	nw vle8 seedLUT x2	Newton	28.6k	478us	65	pv vle8 Vb x2	PV pre	27.9k	466us
2	im2col vle8 idxH2 x2	im2col	27.6k	462us	18	QK vwredsumu.vs x512	QK x512	5933.6k	99,310us	41	nw vrgather seed8 x2	Newton	84.4k	1,409us	66	pv vmv 0acc0 x2	PV pre	27.3k	457us
3	im2col vle8 src x2	im2col	27.6k	462us	19	QK vnsrl->>S x512	QK x512	2529.3k	42,570us	42	nw vwmulu seed16 x2	Newton	9.1k	153us	67	PV vrgather Va col-d x128	PV x128	1848.2k	30,907us
4	im2col vrgather H1 x2	im2col	44.4k	742us	20	QK vminu255 x512	QK x512	1881.4k	31,769us	43	nw vwmulu R0_32 x2	Newton	9.1k	153us	68	PV vwmulu pqa*Va x128	PV x128	815.2k	13,689us
5	im2col vrgather V x2	im2col	84.5k	1,409us	21	QK vnsrl->u8 x512	QK x512	2528.9k	42,562us	44	nw vmul i1 x2	Newton	7.8k	132us	69	PV vwredsumu accA x128	PV x128	2269.1k	37,922us
6	im2col vrgather H2 x2	im2col	63.7k	1,064us	22	QK vslideup x512	QK x512	6342.8k	106,130us	45	nw vrsub i1 x2	Newton	7.4k	124us	70	PV vrgather Vb col-d x128	PV x128	1848.1k	30,905us
7	im2col vse8 T x2	im2col	25.7k	431us	23	QK vmerge x512	QK x512	7315.8k	122,347us	46	nw vmul i1 x2	Newton	7.8k	132us	71	PV vwmulu pqb*Vb x128	PV x128	815.3k	13,691us
8	transpose vle8(V) x2	transpose	27.2k	454us	24	QK vmv x512	QK x512	7052.5k	117,960us	47	nw vsrl i1 x2	Newton	7.8k	132us	72	PV vwredsumu chain x128	PV x128	2269.1k	37,922us
9	transpose vse8(H) x2	transpose	26.3k	441us	25	attn vmv COPYSA x2	softmax	27.6k	461us	48	nw vmul i2 x2	Newton	7.9k	133us	73	PV vnsrl->>0 x128	PV x128	632.7k	10,649us
10	attn vle8 Q x2	QK pre	27.6k	462us	26	sm vredmaxu Sa x2	softmax	18.6k	312us	49	nw vrsub i2 x2	Newton	7.3k	124us	74	PV vminu255 x128	PV x128	470.3k	7,941us
11	attn vid x2	QK pre	26.0k	435us	27	sm vredmaxu both x2	softmax	18.7k	313us	50	nw vmul i2 x2	Newton	7.8k	132us	75	PV vnsrl->u8 x128	PV x128	632.1k	10,639us
12	attn vmseq x2	QK pre	11.1k	187us	28	sm vrgather m x2	softmax	28.9k	483us	51	nw vsrl i2 x2	Newton	7.8k	132us	76	PV vslideup x128	PV x128	1063.3k	17,825us
13	attn vmv0 x2	QK pre	27.3k	457us	29	sm vle8 expLUT x2	softmax	29.0k	485us	52	nw vmul i3 x2	Newton	7.8k	132us	77	PV vmerge x128	PV x128	1172.8k	19,650us
14	attn vle8 Kt x4	QK pre	55.6k	930us	30	sm vssubu m-Sa x2	softmax	9.4k	159us	53	nw vrsub i3 x2	Newton	7.3k	124us	78	PV vmv x128	PV x128	1105.9k	18,536us
15	attn vmv sczero x4	QK pre	54.6k	913us	31	sm vminu127 x2	softmax	9.1k	153us	54	nw vmul i3 x2	Newton	7.8k	132us	79	attn vse8 0 x2	store	23.6k	394us
					32	sm vrgather ea x2	softmax	49.4k	826us	55	nw vsrl i3 x2	Newton	7.8k	132us					
					33	sm vssubu m-Sb x2	softmax	9.4k	158us	56	norm vnsrl R16 x2	norm	9.9k	166us					
					34	sm vminu127 x2	softmax	9.1k	153us	57	norm vrgather Rb16 x2	norm	33.0k	551us					
					35	sm vrgather eb x2	softmax	52.1k	870us	58	norm vwmulu e16a x2	norm	12.5k	210us					
					36	sm vwredsumu Z(a) x2	softmax	24.0k	401us	59	norm vmul P16a x2	norm	12.9k	217us					
					37	sm vwredsumu Z(b) x2	softmax	24.0k	402us	60	norm vnsrl pqa x2	norm	32.7k	547us					
					38	sm vwmulu Z32 x2	softmax	9.1k	153us	61	norm vwmulu e16b x2	norm	12.5k	210us					
										62	norm vmul P16b x2	norm	12.9k	217us					
										63	norm vnsrl pqb x2	norm	32.7k	547us					